Samples

- HS1
- HS2
- HS3
- HS4
- HS5
- HS6
- HS7
- HS8
- HS9

TOY EXPERIMENT ON 3×3 CHECKERBOARD CLASSIFICATION DATASET

\# Neurons = 4-4
\# Activation Patterns = 6
\# Samples = 1350 (train) 675 (test)
\# Epochs = 20000
Learning Rate = 0.001
Train Accuracy = 0.9830
Test Accuracy = 0.9837
Seed = 13112

Train samples

Pattern	#Samples	TP	FP	TN	FN
0001-0011	68	68	0	0	0
1001-0011	234	81	1	151	1
1011-0000	176	0	0	176	0
1011-0010	274	175	3	91	5
1011-0011	418	265	4	144	5
1011-0100	180	148	2	28	2

Test samples

Pattern	#Samples	TP	FP	TN	FN
0001-0011	29	29	0	0	0
1001-0011	125	46	3	76	0
1011-0000	82	0	0	82	0
1011-0010	130	83	3	44	0
1011-0011	213	141	3	68	1
1011-0100	96	75	1	20	0

ACTIVATION PATTERN INEQUALITIES

$0.0000 \leq x1 \leq 1.0000$
$0.0000 \leq x2 \leq 1.0000$
N=1, HS1: $x1 - 1.0181x2 + 0.6898 \geq 0$
N=0, HS2: $x1 - 0.7216x2 - 1.7434 \leq 0$
N=1, HS3: $x1 - 0.9940x2 + 0.3244 \geq 0$
N=1, HS4: $x1 - 0.9288x2 - 0.9714 \leq 0$
N=0, HS5: $x1 - 0.7723x2 - 4.8990 \leq 0$
N=0, HS6: $x1 - 0.9737x2 - 0.4294 \leq 0$
N=0, HS7: $x1 - 0.9694x2 - 0.3102 \leq 0$
N=0, HS8: $x1 - 0.9789x2 - 0.0246 \leq 0$
$Y\_r \geq Y\_g$, HS9: $x1 - 0.9789x2 - 0.0246 \leq 0$

RULE ANTECEDENTS

- **Class R (red)**
  No Apply

- **Class G (green)**
  $x2 \geq 0$
  $x1 - 1.0000 \leq 0$
  $x1 - 0.9737x2 - 0.4294 \leq 0$
  $x1 - 0.9694x2 - 0.3102 \geq 0$

RULE CONSEQUENT (Network Output)

$Y\_r = -9.0974$
$Y\_g = 9.0883$

ACTIVATION REGION (Classes R&G)

$x2 \geq 0$
$x1 - 1.0000 \leq 0$
$x1 - 0.9737x2 - 0.4294 \leq 0$
$x1 - 0.9694x2 - 0.3102 \geq 0$